Ignorance Before Uncertainty

The Structure, Geometry, Classification and Hard Limits of Uncertainty in Resolution Geometry

Matěj Rada
MatejRada@email.cz
Licensed under CC BY-NC-ND 4.0

November 7, 2026 (print edition)

Abstract

We settle the place of uncertainty in RG.

The hierarchy. *Ignorance* is a fibre: the descriptions that a declared protocol cannot separate. It needs only records and equality. *Uncertainty* is ignorance at finite resolution, together with a declared quantification. Two further levels add more structure:

- probabilistic uncertainty additionally needs stable frequencies and repetition;
- geometric uncertainty additionally needs smoothness and regularity.

Each level adds a separable assumption.

Ignorance is the more fundamental notion.

(i) Each can occur without the other: a deterministic record x^2 has a fibre and zero spread, while an identifiable coin has spread and no fibre.

(ii) Ignorance is the residue of uncertainty that no amount of data removes. Worst-case error stays at least the fibre half-diameter at every sample size, while the statistical part vanishes.

So RG's order (ignorance first) is correct, and uncertainty is not its cause.

Uncertainty is not identical to probability. Set-valued uncertainty exists without it. Probability is the quantification that stable repeated records support. Where they do not (single events, non-stationary processes), a probability is a declaration.

The canonical spread. The canonical object of spread is the *posterior law* under convex order, and it is unique up to experiment equivalence. Scalar spreads (variance, entropy) are task functionals: two experiments can have equal expected posterior variance and different posterior laws. Monte Carlo is a sampler of laws, not a deeper spread.

Geometry. Uncertainty does *not* deserve an independent geometry. Its local geometry is the dual of the forced distinguishability geometry: efficient covariance is F^{-1}, transforming contravariantly in every chart. Its global geometry is the confirmation topology together with Le Cam's distance. The uncertainty ellipsoids we drew are correct, but they are derived pictures.

Hard limits. We prove a catalogue of floors:

- the fibre floor;
- the two-point (global) floor;
- the Cramér–Rao (local) floor and its nuisance form;

- singular-wall rates ($n^{-1/4}$ where Cramér–Rao is void);
- the Fano capacity floor;
- the resolution ratio;
- the confirmation floor;
- the distribution-free $n^{-1/2}$ width.

Quantum uncertainty, identified. Three distinct kinds are each derived from positivity on a noncommuting algebra:

- *preparation:* Robertson from Cauchy–Schwarz; for a qubit, $\operatorname{Var} X + \operatorname{Var} Z \geq 1$ and $H(X) + H(Z) \geq 1$ bit;
- *joint measurement:* $\eta_x^2 + \eta_z^2 \leq 1$;
- *estimation:* quantum Fisher information N for product states and N^2 for GHZ states, a $\sqrt{N}$ ratio in error.

Commuting observables admit dispersion-free joint states. Quantum uncertainty is therefore a positivity floor, not a hidden value left unread.

Luck. Luck is a property of a realization, uncertainty of a law, and ignorance of a protocol. Outcomes are shaped by uncertainty exactly where fibres are nontrivial or records are stochastic.

Contents

1 The questions and their answers **2**

2 A hierarchy of structure **3**

3 Ignorance is the more fundamental notion **3**

4 Is uncertainty probability? **4**

5 The canonical spread **4**

6 Does uncertainty deserve its own geometry? **4**

7 Hard limits: the classical catalogue **5**

8 Quantum uncertainty, identified **5**

9 Luck, and which outcomes uncertainty shapes **6**

10 Final ledger **6**

11 Reproduction and limits **7**

1 The questions and their answers

Question	Answer (proved below)
Which is more universal, ignorance or uncertainty?	Ignorance. It needs fewer assumptions, and it is the infinite-data residue of uncertainty (Theorems 3.1, 3.2).
Does uncertainty cause ignorance, or the reverse?	Neither. They are independent dimensions, and ignorance is the floor that uncertainty converges to.

Question	Answer (proved below)
Does uncertainty deserve its own geometry?	No. Locally it is the dual of Fisher; globally it is the confirmation topology and Le Cam (Section 6).
Is uncertainty probability?	No. Probability is one quantification, available under stable repetition (Section 4).
Is Monte Carlo the deepest spread?	No. The posterior law is the canonical spread, and Monte Carlo samples it (Section 5).
What are the hard limits?	Eight classical floors and three quantum ones (Sections 7–8).
How does it differ from luck?	Luck belongs to a realization, uncertainty to a law, ignorance to a protocol (Section 9).
Are all outcomes shaped by uncertainty?	Only where fibres are nontrivial or records are stochastic. Exact deduction is not.
Is RG written in the right order?	Yes. Starting from uncertainty would need more assumptions.

2 A hierarchy of structure

Definition 2.1 (The levels)**.** (L0) *Ignorance.* For a profile map Φ (exact record laws or exact records), the fibre $[\theta]$, and for a task Q its fibre range $Q([\theta])$. The only structure needed is equality of records.

(L1) *Uncertainty.* Ignorance at finite resolution, with a declared quantification. This may be set-valued (the set of descriptions compatible with the record up to a tolerance, with a metric on Q) or a measure.

(L2) *Probabilistic uncertainty.* Level L1 together with stable frequencies (Assumption S of *From the Wall-Tap*) and repetition (Assumption I).

(L3) *Geometric uncertainty.* Level L2 together with smooth regular models. Its local form is Fisher and its dual.

(L4) *Quantum uncertainty.* Level L3 on a noncommutative algebra of observables, with states that are positive functionals.

Proposition 2.2 (The hierarchy is strict)**.** *Each level adds an assumption that can fail while the lower level holds:*

- *a deterministic exact record has ignorance but no quantification (L0 without L1);*
- *a tolerance set with no frequencies is L1 without L2;*
- *a non-smooth or singular model is L2 without L3;*
- *a commutative algebra is L3 without L4.*

3 Ignorance is the more fundamental notion

Theorem 3.1 (Independence)**.** *Ignorance and uncertainty can each occur without the other.*

Proof. Ignorance without uncertainty (U1). The deterministic record $x \mapsto x^2$ has zero spread and the nontrivial fibre $\{x, -x\}$.

Uncertainty without ignorance (U2). Bernoulli(θ) is identifiable, with spread $\theta(1-\theta)/n \to 0$. □

Theorem 3.2 (Decomposition: ignorance is the residue)**.** *Let Q be a task with values in a metric space, and let $D(\theta) = \sup\{d(Q(\theta'), Q(\theta'')) : \theta', \theta'' \in [\theta]\}$ be its fibre diameter.*

(a) For every sample size n and every estimator T, $\sup_{\theta' \in [\theta]} \mathbb{E}_{\theta'} d(T, Q(\theta')) \geq D(\theta)/2$.

(b) Suppose the profile is consistently estimable, Q is bounded on fibres, and the fibre range has a centre c with $d(c, Q(\theta')) \leq D/2$ (for example, real-valued Q). Then a rule that estimates the profile and outputs the centre of the fibre range has worst-case error tending to $D(\theta)/2$.

Total uncertainty is therefore a statistical part, which vanishes with data, on top of an ignorance floor, which does not.

Proof. (a) All members of the fibre give the same record law, so T has one law across the fibre. By the triangle inequality, $\mathbb{E}d(T, Q(\theta')) + \mathbb{E}d(T, Q(\theta'')) \geq d(Q(\theta'), Q(\theta''))$.

(b) The profile estimate converges, so with probability tending to 1 the selected fibre is the true one, and the centre is within $D/2$.

Witness (U3): for records of Bernoulli(θ^2) and the task $Q = \theta$, the floor is $|\theta| = 0.4$ at $n = 10, 100, 1000$. The statistical part of $|\theta|$ has standard deviation 0.145, 0.046 and 0.0145. □

Corollary 3.3 (RG is written in the right order)**.** *Ignorance needs only records and equality. Uncertainty adds a quantification and, in its probabilistic form, stable repetition. Ignorance is also the limit of uncertainty. Starting from uncertainty would assume more, and would still have to recover the fibre as its limit.*

4 Is uncertainty probability?

Proposition 4.1 (Probability is one quantification)**.**

- *Under stable frequencies, normalization and finite additivity follow from counting (*From the Wall-Tap*, Proposition 2.1). This is the frequency route to probability.*
- *Set-valued uncertainty (tolerance sets, interval bounds, fibre ranges) needs no probability.*
- *Where stable repetition is absent (single events, non-stationary processes), a probability is a* declaration*: a prior or a model, typed as such.*
- *Coherence routes to probability (Cox; de Finetti) are alternative* declarations *of a quantification. They are imported, not derived from records [1, 2].*

Probability geometry. The geometry of probabilistic uncertainty is the geometry of record laws: Fisher locally, and the confirmation topology and Le Cam distance globally. Set-valued uncertainty carries only the metric and topology of the task space. Neither is a geometry *of uncertainty* independent of the laws.

5 The canonical spread

Theorem 5.1 (The posterior law is the complete spread)**.** *For a finite parameter set and a full-support prior, the posterior law μ_E (the distribution of posterior beliefs) determines the experiment up to equivalence. Every operational spread functional is monotone in its convex order (*Experiments, Simulation and Descent*). A scalar spread is a task functional and is not complete.*

Witness (U8).

- Experiment A has posteriors $\{0.3, 0.7\}$ with masses $\frac{1}{2}, \frac{1}{2}$.

- Experiment B has posteriors $\{0.5, 0.9, 0.1\}$ with masses $0.75, 0.125, 0.125$.

Both have the same expected posterior variance of θ, 0.0336, but different posterior laws, so they are not equivalent.

Proposition 5.2 (Monte Carlo is a sampler)**.** *Monte Carlo estimates of a law's functionals converge at rate $n^{-1/2}$. The Chebyshev width from* From the Wall-Tap*, Lemma 5.2, is distribution-free. Monte Carlo spread is the sampling spread of an estimator of the law. It is determined by the law and the estimator, and is not a deeper object than the law.*

6 Does uncertainty deserve its own geometry?

Theorem 6.1 (Uncertainty geometry is the dual of the forced geometry)**.** *For a regular model, the efficient covariance is F^{-1}. Under a change of chart $\theta = \phi(\eta)$ with Jacobian J,*

$$F_\eta = J^\top F_\theta J, \qquad \Sigma_\eta = J^{-1}\Sigma_\theta J^{-\top},$$

so $\Sigma = F^{-1}$ holds in every chart (U4, exact). By duality, any positive quadratic uncertainty form on cotangent vectors that is invariant under lossless splittings is $c^{-1}F^{-1}$.

Proof. Transform the score: $s_\eta = J^\top s_\theta$. Invariance of the dual form is the invariance of its inverse, and the inverse is forced to be $c \cdot F$ by the forced-geometry theorem of *From the Wall-Tap*. □

Answer. Uncertainty is bound to the distinguishability geometry. Locally its ellipsoids are F^{-1}. Globally its regions are neighbourhoods of the confirmation topology, sized by Le Cam's distance. Drawing uncertainty ellipsoids is justified as the dual picture. Treating them as an independent geometry is not justified.

7 Hard limits: the classical catalogue

Floor	Statement	Proof / witness
Fibre floor	worst-case error $\geq D/2$ at every n	Theorem 3.2; U3
Two-point (global)	$\inf_T \max_i \mathbb{E}_i\lvert T - Q_i\rvert \geq \frac{\Delta}{2}(1 - \mathrm{TV}(P_1^n, P_2^n))$	testing identity; hybrid bound $\mathrm{TV}(P^n, Q^n) \leq n\,\mathrm{TV}$
Cramér–Rao (local)	$\mathrm{Var}\,T \geq F^{-1}$; with nuisance, $(F^{-1})_{11} = (F_{11} - F_{12}F_{22}^{-1}F_{21})^{-1}$	score Cauchy–Schwarz; Schur complement
Singular wall	where $F = 0$, the rate is $n^{-\lambda}$, e.g. $n^{-1/4}$ for $p = (\frac{1}{2} \pm t^2)$	U5: TV $= t^2$; Rooting Closure (RLCT)
Fano (capacity)	$P_{\mathrm{err}} \geq 1 - (k+1)/\log_2 M$ for k bits and M hypotheses	U6: 0.45 for $M = 2^{20}$, $k = 10$
Resolution ratio	two frequencies $\Delta\omega$ apart need $T \gtrsim 2\pi/\Delta\omega$	Gram conditioning (Tides: 6.9 at 29 days, 1.0 at a year for K1/P1)
Confirmation	optimal error $\frac{1}{2}$ before the mediated arrival	Wall-Tap Structural
Distribution-free width	$\lvert f_n - p\rvert < (4n\alpha)^{-1/2}$ with probability $\geq 1 - \alpha$	Chebyshev

Ratios.

- Efficiency is $\mathrm{Var}\,T \cdot F \geq 1$, with equality for efficient estimators.

- Global over local: the two-point floor can be positive where Cramér–Rao is void (U5).
- Rate exponents: $\frac{1}{2}$ (regular) against $\lambda < \frac{1}{2}$ (singular) per parameter.

8 Quantum uncertainty, identified

Theorem 8.1 (Preparation floor from positivity)**.** *For self-adjoint A, B and any state,*

$$\operatorname{Var} A \cdot \operatorname{Var} B \geq \tfrac{1}{4}|\langle [A, B]\rangle|^2.$$

If A and B commute, there are joint eigenstates, which are dispersion-free in both.

Proof. Put $a = A - \langle A\rangle$ and $b = B - \langle B\rangle$. By Cauchy–Schwarz for the positive form $(X, Y) \mapsto \langle X^\dagger Y\rangle$, $\langle a^2\rangle\langle b^2\rangle \geq |\langle ab\rangle|^2 \geq (\operatorname{Im}\langle ab\rangle)^2 = \frac{1}{4}|\langle [A, B]\rangle|^2$. Commuting self-adjoint matrices are simultaneously diagonalizable. □

Qubit (U7). On the Bloch ball:

- $\operatorname{Var} X + \operatorname{Var} Z = 2 - r_1^2 - r_3^2 \geq 1$;
- $\operatorname{Var} X \operatorname{Var} Y - \langle Z\rangle^2 \geq r_1^2 r_2^2 \geq 0$;
- $H(X) + H(Z) \geq 1$ bit, the Maassen–Uffink bound [3] (grid minimum 1.000).

Imported Theorem 8.2 (Joint measurement and estimation [4, 5])**.** *Unbiased joint approximations of σ_x and σ_z require $\eta_x^2 + \eta_z^2 \leq 1$. The quantum Cramér–Rao bound is* $\operatorname{Var} \geq 1/(nF_Q)$.

Estimation (U7, exact). For phase estimation, the quantum Fisher information is N for product states (the standard limit) and N^2 for GHZ states (the Heisenberg limit), for $N = 1, \ldots, 6$. The error ratio is $\sqrt{N}$.

Corollary 8.3 (Identification)**.** *Quantum uncertainty is three distinct floors, each derived from positivity on a noncommuting algebra:*

- preparation*, with no dispersion-free state for a noncommuting pair;*
- joint measurement*, with sharpness traded off;*
- estimation*, with quantum Fisher information as the resource.*

It is not fibre ignorance of an unread hidden value. The Kochen–Specker and Bell theorems exclude such a reading under noncontextuality and locality [6, 7]. In RG typing it is a complementarity floor*, which a commutative algebra lacks.*

9 Luck, and which outcomes uncertainty shapes

Definition 9.1 (Luck)**.** For a realized record x and a decision rule with loss ℓ, the luck of the realization is $\mathbb{E}\ell - \ell(x)$.

Proposition 9.2. *Luck has mean zero, and its spread is the uncertainty of the law. Luck is not predictable from the records: if it were, it would be part of the law. In short, luck is a property of a realization, uncertainty of a law, and ignorance of a protocol.*

Proposition 9.3 (Where uncertainty acts)**.** *A record is shaped by uncertainty iff its protocol has a nontrivial fibre for the task or stochastic records.*

- *Exact deduction and exact counting have neither.*
- *Floating-point arithmetic has both, as a protocol of finite resolution.*
- *Physics has stochastic records wherever repetition does not give identical outcomes, and a quantum floor wherever noncommuting observables are both read.*

10 Final ledger

Claim	Status	Basis
Ignorance and uncertainty are independent	THEOREM	U1, U2
Ignorance is the infinite-data residue of uncertainty	THEOREM	Theorem 3.2, U3
Uncertainty $\neq$ probability; probability needs stable repetition	THEOREM + IMPORTED	Wall-Tap; Cox, de Finetti
The canonical spread is the posterior law; scalars are task functionals	THEOREM + WITNESS	U8
Uncertainty geometry is the dual of Fisher, not standalone	THEOREM	Theorem 6.1, U4
Eight classical floors	THEOREMS	Section 7
Three quantum floors from positivity on a noncommuting algebra	THEOREM + IMPORTED	Theorem 8.1, U7
Luck vs uncertainty vs ignorance	DEFINITION + PROPOSITION	Section 9
RG's order (ignorance first)	CONFIRMED	Corollary 3.3

11 Reproduction and limits

- `rg_uncertainty_structure.py` runs U1–U8. The witnesses are exact, symbolic, or a fixed grid (the Maassen–Uffink minimum). Reruns are byte-identical.
- *Limits.*
  - The decomposition theorem's attainability needs consistent profile estimation and a bounded task.
  - Entropic and joint-measurement bounds are imported; the Maassen–Uffink check is a grid on pure qubit states.
  - The quantum identification is for finite dimensions.

References

[1] R. T. Cox, Probability, frequency and reasonable expectation, *Am. J. Phys.* 14 (1946) 1.

[2] B. de Finetti, La prévision: ses lois logiques, ses sources subjectives, *Ann. Inst. H. Poincaré* 7 (1937) 1.

[3] H. Maassen, J. B. M. Uffink, Generalized entropic uncertainty relations, *Phys. Rev. Lett.* 60 (1988) 1103.

[4] P. Busch, Unsharp reality and joint measurements for spin observables, *Phys. Rev. D* 33 (1986) 2253.

[5] S. L. Braunstein, C. M. Caves, Statistical distance and the geometry of quantum states, *Phys. Rev. Lett.* 72 (1994) 3439.

[6] S. Kochen, E. P. Specker, The problem of hidden variables in quantum mechanics, *J. Math. Mech.* 17 (1967) 59.

[7] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics* 1 (1964) 195.

[8] M. Rada, *Resolution Geometry* (print edition) and companions: From the Wall-Tap; Experiments, Simulation and Descent; Rooting Closure; Wall-Tap Structural; Tides; One World, 2026.